

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import urllib.request

# -----
# INPUT
# -----
url = "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/butterfly.jpg"
urllib.request.urlretrieve(url, "input.jpg")

img = cv2.imread("input.jpg")

if img is None:
    raise Exception("Image could not be loaded")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# -----
# SIFT FEATURE DETECTOR
# -----
sift = cv2.SIFT_create()

# Different blur levels
kernels = [1, 3, 5, 9, 15]

feature_counts = []

plt.figure(figsize=(15, 8))

for i, k in enumerate(kernels):

    if k == 1:
        blurred = img.copy()
    else:
        blurred = cv2.GaussianBlur(img, (k, k), 0)

    gray_blur = cv2.cvtColor(blurred, cv2.COLOR_BGR2GRAY)

    keypoints, descriptors = sift.detectAndCompute(
        gray_blur,
        None
    )

    count = len(keypoints)
    feature_counts.append(count)

    output = cv2.drawKeypoints(
        blurred,
        keypoints,
        None,
        flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
    )

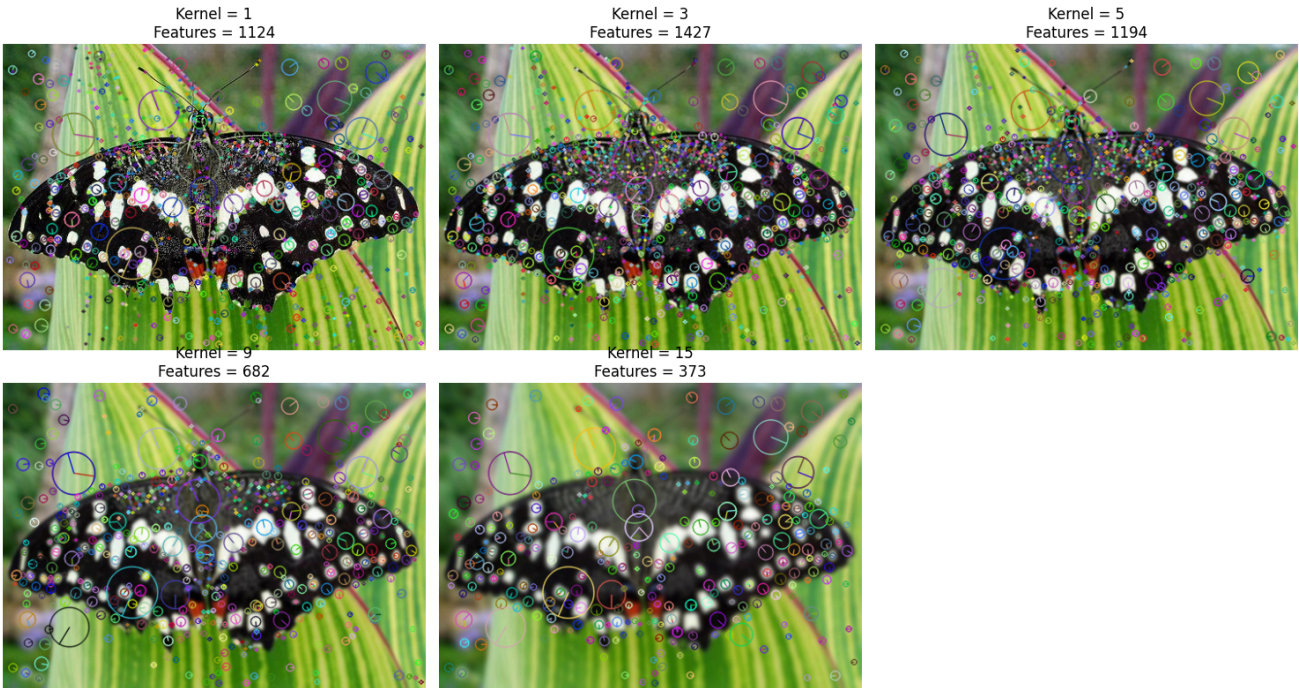
    plt.subplot(2, 3, i + 1)
    plt.imshow(cv2.cvtColor(output, cv2.COLOR_BGR2RGB))
    plt.title(f"Kernel = {k}\nFeatures = {count}")
    plt.axis("off")

plt.tight_layout()
plt.savefig("Q27_blur_features.png")
plt.show()

# -----
# PRINT RESULTS
# -----
print("\nEFFECT OF BLURRING ON FEATURE DETECTION")
print("-" * 50)

for k, count in zip(kernels, feature_counts):
    print(f"Blur Kernel {k}x{k} : {count} features")

# -----
# GRAPH
# -----
plt.figure(figsize=(8, 5))
plt.plot(kernels, feature_counts, marker="o")
plt.xlabel("Gaussian Blur Kernel Size")
plt.ylabel("Number of Detected Features")
plt.title("Effect of Blurring on Feature Detection")
plt.grid(True)
plt.savefig("Q27_blur_graph.png")
plt.show()
```



EFFECT OF BLURRING ON FEATURE DETECTION  
-----  
Blur Kernel 1x1 : 1124 features  
Blur Kernel 3x3 : 1427 features  
Blur Kernel 5x5 : 1194 features  
Blur Kernel 9x9 : 682 features  
Blur Kernel 15x15 : 373 features

